

The best nuclear stock you've never heard of

Hey Disruption Investors,

My partner Chris Wood and I pride ourselves on keeping you ahead of the curve. It's our job to help you see around corners and make money from disruption before it's on the front page of *The Wall Street Journal*.

One trend we were very early to was the nuclear energy renaissance.

We're approaching the seven-year anniversary of our first writeup on uranium (nuclear fuel) producer **Cameco Corp. (CCJ)**. Our only mistake was that we weren't bullish enough.

Cameco's stock has surged 7X since then:



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How We'll Flip a Bloodbath into 3x Gains



STEPHEN MCBRIDE * AUGUST 2, 2018

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Source: RiskHedge

In this month's issue

When most people read the world "nuclear," they think "bombs"

ChatGPT sparked an energy revolution

AI is nuclear's best friend

The best business in nuclear you've never heard of

Key prices

S&P 500	6,192.45
1-Month (Gain)	+3.58%
Nasdaq	20,206.62
1-Month (Gain)	+3.91%
Russell 2000	2,197.54
1-Month (Gain)	+4.48%
Dollar Index	97.05
1-Month (Loss)	-2.19%
US 10-Year Yield	4.295%



A RiskHedge Publication

For the seven years I've been writing about nuclear, the story has been one of grim survival. The investment case was, "I ran the numbers, and it looks like we won't shut down as many plants as expected."

Now, the world is making a 180-degree turn on atomic energy. Few investors grasp the speed of this shift. But after decades of stagnation, we're entering a nuclear renaissance.

And this month, we've found the best way to profit from this historic U-turn...

When most people read the word “nuclear,” they think “bombs”

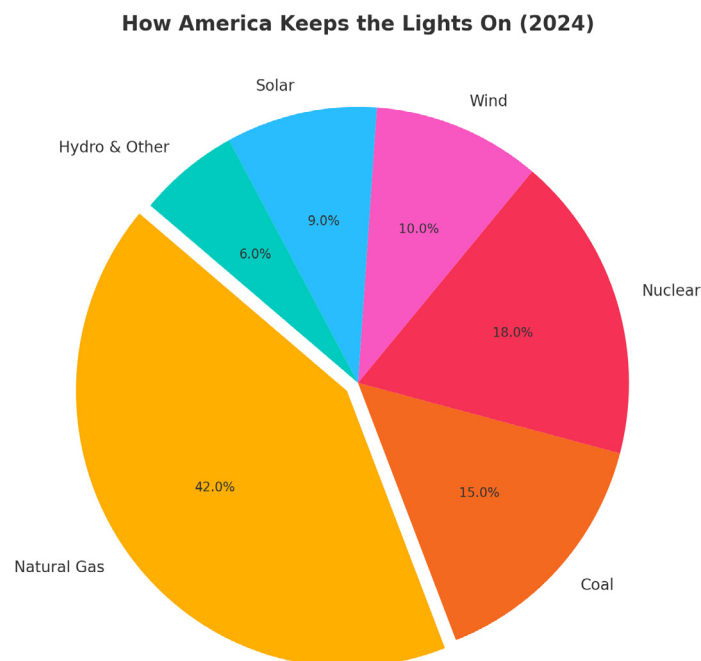
Atomic technology was used to make the most destructive weapons in human history.

But modern uses of nuclear power are overwhelmingly good.

It's hard to think of a tougher assignment for a PR pro than painting nuclear power in a positive light. If you've ever watched *The Simpsons*, you'll know evil Mr. Burns owned the nuclear power plant. There, Homer Simpson nods off in the control room and toxic sludge pours into the local river.

Just like *The Simpsons*, that scenario is fiction. Nuclear isn't dirty or dangerous.

For starters, nuclear power (red) keeps the lights on for about one in five American homes today:

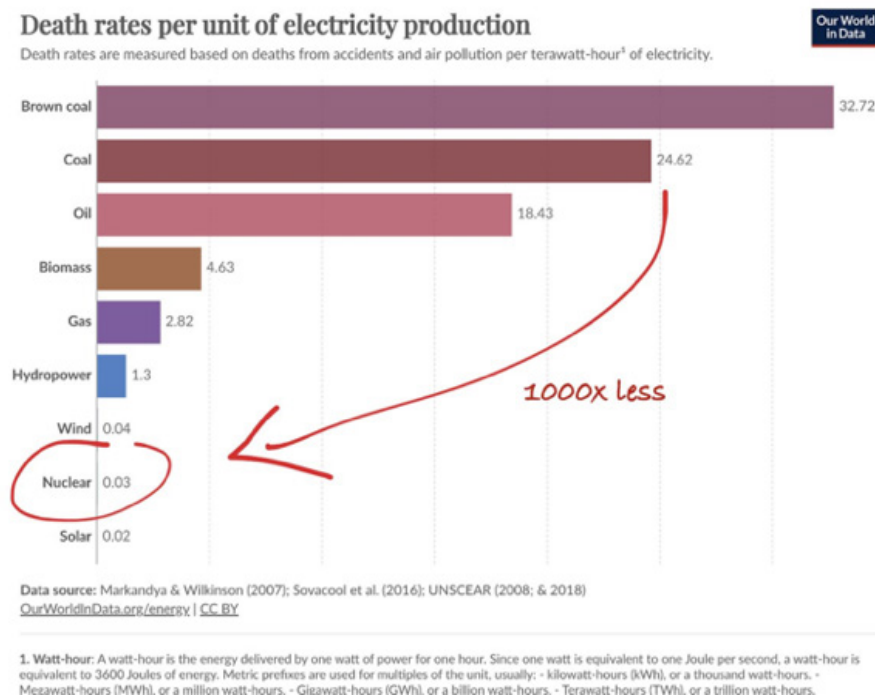


Source: Created by ChatGPT

I think we should power 100% of America's energy needs with nuclear. It's the only energy source that scores a perfect four: clean, safe, dense, and reliable.

Let's walk through each point below.

Safe: For every unit of energy produced, brown coal kills over 1,000X more people than nuclear. It's just as safe as solar. In fact, you receive more radiation from eating a single banana than you do from living next to a nuclear power plant for a year.



Source: Our World in Data

"But what about Fukushima, Three Mile Island, and Chernobyl?"

I'm glad you asked. All three nuclear "accidents" are misunderstood.

Fukushima: No one died from the reactor explosion or the nuclear release. Only one radiation-related death has been reported. Today, the natural background radiation in high-altitude places, like Colorado, is higher than in Fukushima.

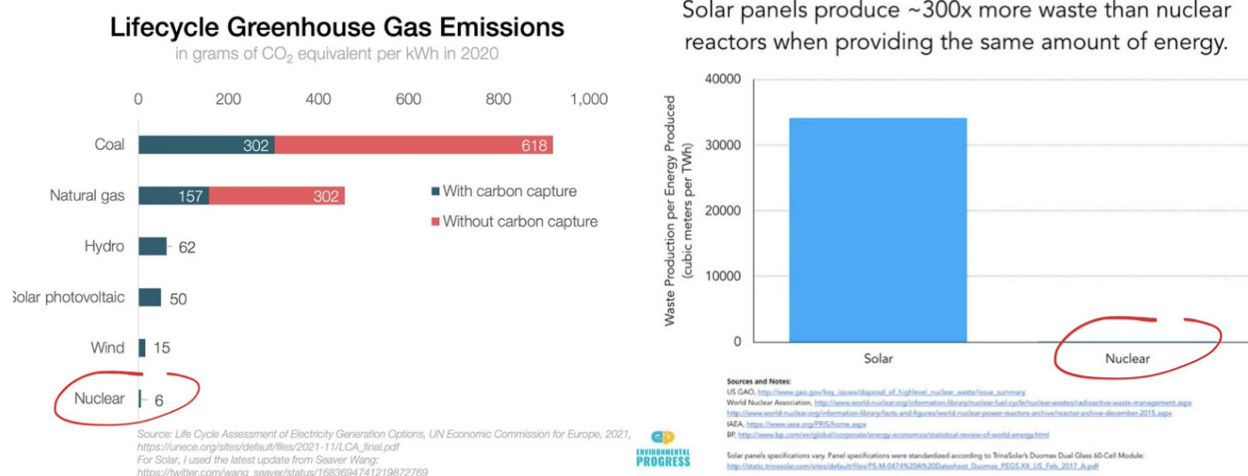
Three Mile Island: No one died in the reactor failure, nor have radiation-related deaths been found.

Chernobyl: A true disaster that was possible because the Soviet reactor's design included positive feedback loops that allowed a runaway reaction when combined with operator error. Today's reactors use negative feedback to shut themselves down. Every death was preventable.

Even including the Chernobyl tragedy, nuclear is still tied for the safest energy source.

Clean: When accounting for the entire lifecycle, from mining to waste disposal, nuclear power emits less CO₂ than any other source. And its waste is minuscule. All the US's nuclear waste produced over the past 60+ years could fit on a single football field, stacked less than 10 yards deep.

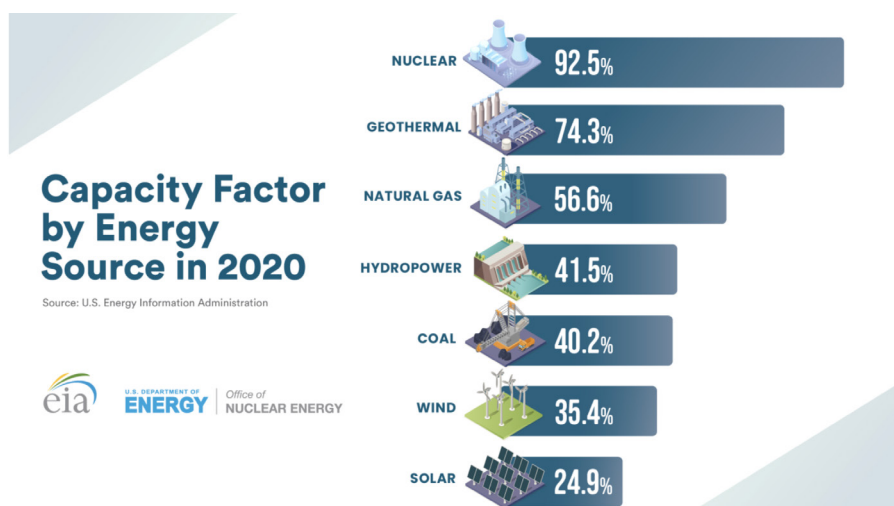
Solar panels, by contrast, produce 300X more waste by volume for the same amount of energy.



Sources: United Nations Economic Commission for Europe; Environmental Progress

Reliable: While solar and wind depend on the whims of the weather, nuclear plants run 24/7, rain or shine, with no fuel price volatility, and with a capacity factor of 92%. That means reactors are producing max power more than 90% of the time during the year.

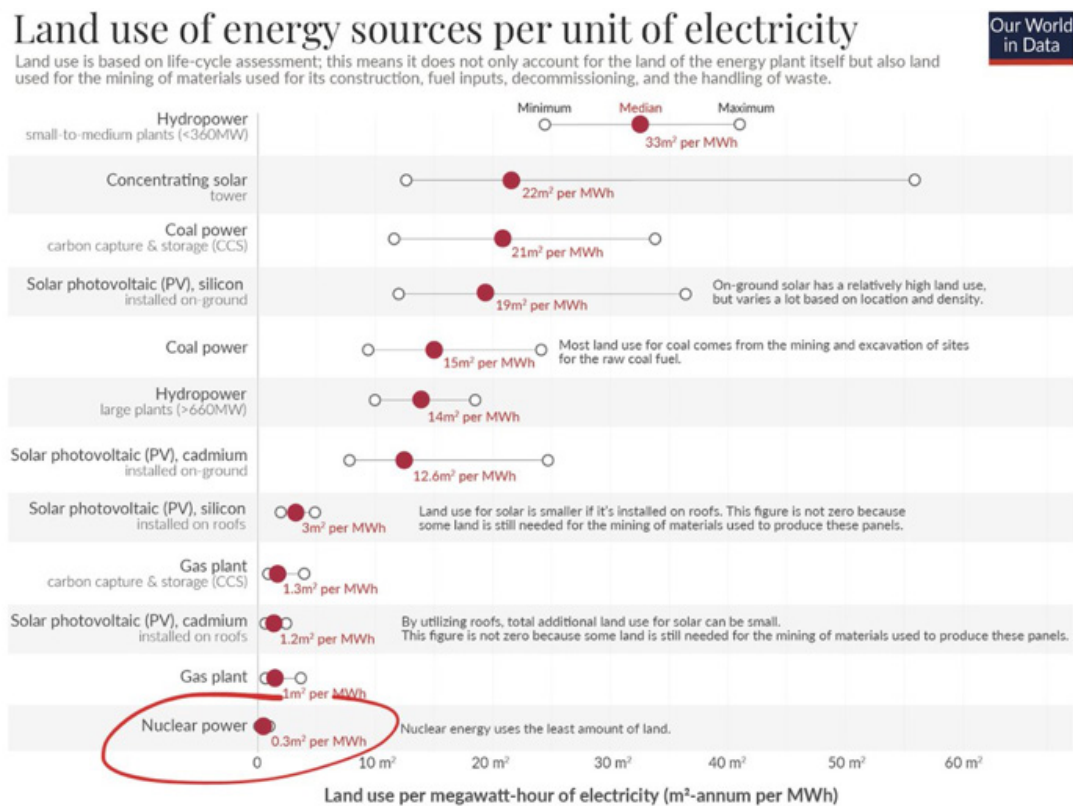
That's 2X more reliable than coal units and 3X more reliable than solar plants:



Source: Seeking Alpha

Dense: A nuclear plant requires a tiny fraction of the land needed for solar or wind farms to produce the same round-the-clock power. A uranium pellet the size of your fingertip holds the same energy as a ton of coal.

It's like stuffing an oil barrel into a sugar cube. If you love pristine, green hills, you should champion nuclear energy. Criticizing nuclear on environmental grounds is backward.



Source: Our World in Data

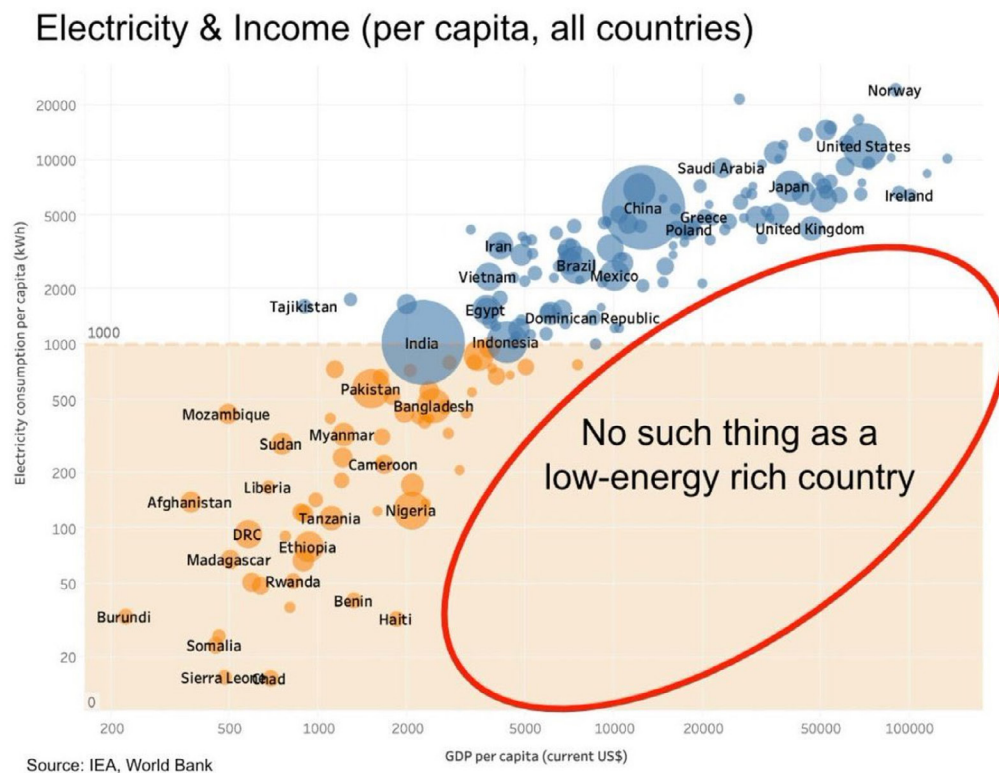
Unfortunately, due to media scaremongering, we turned our back on the best energy source known to man. We've closed more nuclear plants than we've opened this century.

America's nuclear energy output has been flat for 25 years.

This is one of the worst unforced errors America has ever made. Cheap, reliable energy is the bedrock of innovation and prosperity.

Energy is upstream of everything. It's the substrate of civilization. Every breakthrough of the modern world—steel, fertilizer, semiconductors, and space travel—is ultimately a story about energy density. The more energy we have, the more leverage we gain over the material world.

I'll never miss a chance to post one of my all-time favorite charts. *There's no such thing as a low-energy rich country.*



Source: International Energy Agency

Failing to make the leap from fossil fuels to atomic energy caused untold damage over the past half century.

Physical innovation stalled. Climate politics became fear-based instead of solution-based. And America lost its lead in the one energy source that could have underpinned prosperity for generations.

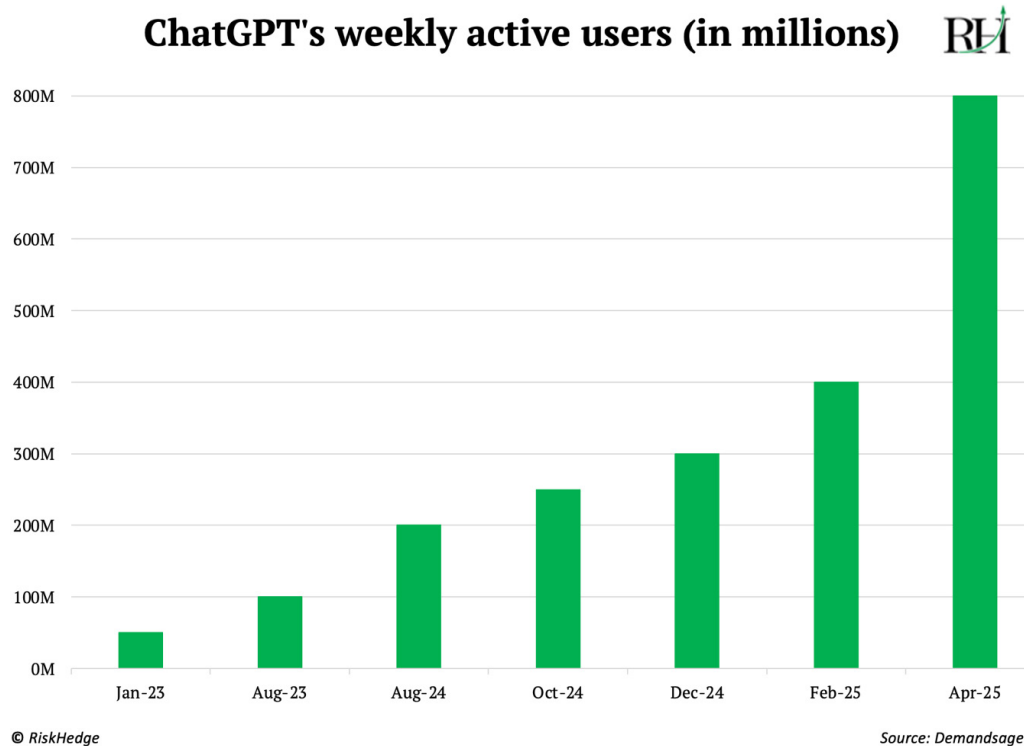
But now, thankfully, we're righting that wrong. And it's all because of another trend Disruption Investors were early to... artificial intelligence (AI).

ChatGPT sparked an energy revolution

When you use ChatGPT, you're tapping into vast data centers filled with specialized chips that run so hot, they need constant cooling just to keep from melting down. One of **Nvidia's (NVDA)** new AI chips draws as much power as two households.

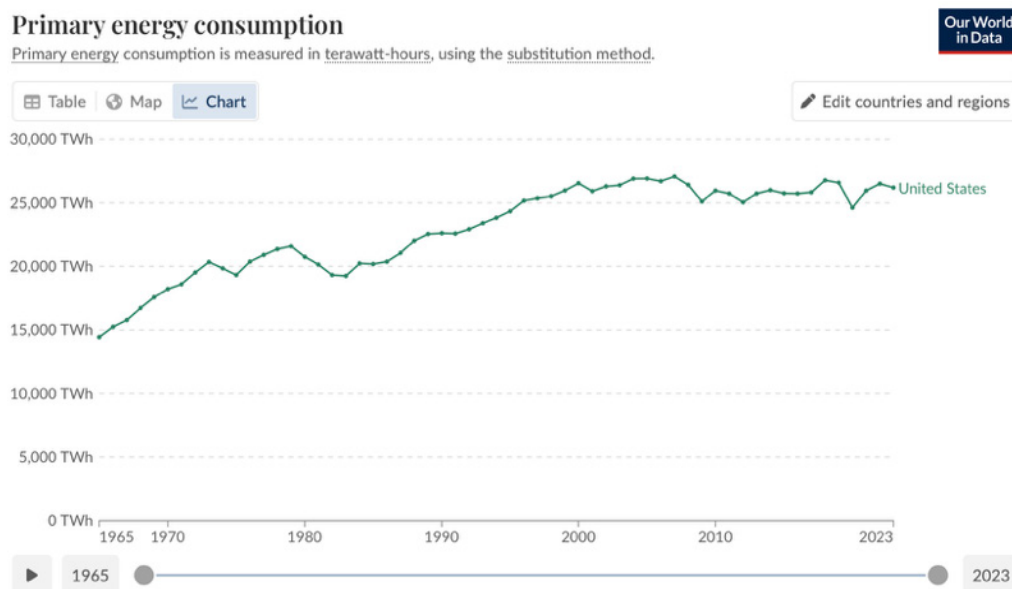
A modern AI data center needs 5X more power than a traditional one. Even a simple ChatGPT query uses *at least* 10X more electricity than a Google search.

ChatGPT launched almost three years ago. It's become the fastest-growing product in history, with roughly 800 million weekly users:



Now, everyone is asking, "How are we going to power all these AI factories?"

Ditching nuclear was a luxury. America's energy consumption has been in a 20-year bear market. "Why do we even need nuclear?" was a legitimate question.



We're entering a new era of energy demand: AI, robotics, reindustrialization, AND electrification all need more power. And not just clean power, but "always on" power. Energy you can bet a \$10 billion data center on.

That, my friends, is forcing everyone to do a U-turn on atomic energy.

Two weeks ago, the World Bank lifted its ban on funding nuclear power projects. The ban has "officially" been in place since 2013. But the last time the bank funded a nuclear project was 1959.

Nuclear plants are expensive to build but cheap to operate. Financing their construction is the key to having many times more power plants.

New York has been the poster child for bad energy policy.

In 2021, it shut down its largest nuclear plant. The closure forced New York to burn more fossil fuels to keep the lights on. More air pollution and higher energy prices, yay!

Thankfully, New York found sense. Last week, Governor Kathy Hochul announced plans to build a new nuclear power plant capable of producing enough electricity for 1 million homes.

Meanwhile, states like Illinois, Montana, and Wisconsin lifted longstanding bans on the construction of new nuclear plants. Texas just approved a \$350 million fund to build new reactors.

State legislatures across America have introduced more than 200 pro-nuclear-related pieces of legislation this year. Now that's what you call an atomic renaissance.

The "coup des gras" for long-suffering nuclear energy cheerleaders like us was President Trump signing four executive orders aimed at freeing nuclear energy from its regulatory shackles last month.

Insane regulations like forcing nuclear plants to reduce radiation exposure to 0.0001 millisieverts per year (less than the amount you ingest when you eat a banana) has made it impossible to build atomic energy.

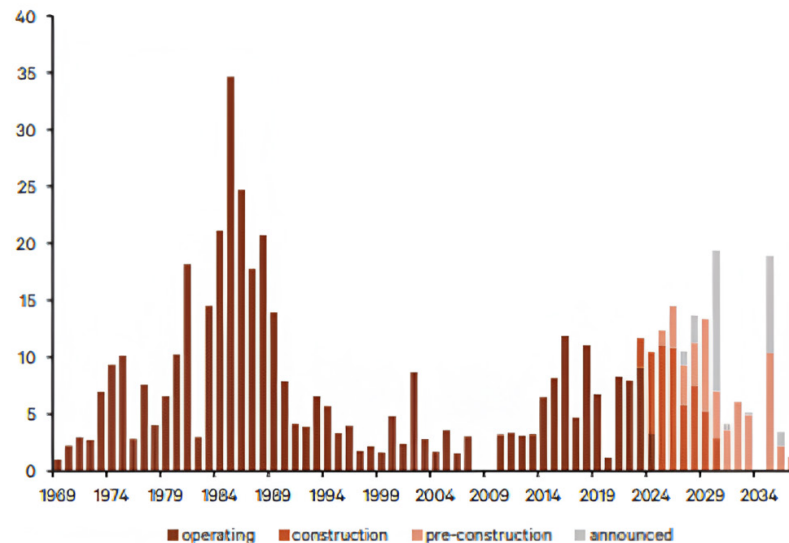
After decades of red tape and bureaucratic nonsense, the dam is finally breaking. AI's insatiable appetite for energy threw a bucket of cold water over regulators and forced them to wake up to the fact we need more clean energy—and now.

It's clear we're past peak nuclear pessimism.

The Palisades plant in Michigan is on track to become the first US nuclear power plant ever restarted. Holtec originally bought the plant to decommission it.

In a dramatic turn, it reversed course and decided to pursue restarting it!

The nuclear energy pipeline is heating up



Global nuclear capacity by start year, GW. Source: Global Energy Monitor

Source: Global Energy Monitor

As the above chart shows, the nuclear pipeline is heating up. And this doesn't even include all the small modular reactors (SMRs) in the works.

AI is nuclear's best friend

AI data centers have an urgent need for massive amounts of clean, reliable energy. Nuclear provides that. Nuclear is to AI what oil was to the industrial age: the fuel for a new era of exponential progress.

Why else do you think big tech companies are sprinting around, checkbooks in hand, hoovering up all the atomic energy they can get their hands on?

For example, in the past year:

- **Microsoft (MSFT)** inked a deal to help restart a reactor at the iconic Three Mile Island plant in Pennsylvania.
- **Google (GOOGL)** signed an agreement with startup Kairos Power to build several SMRs to power its data centers.
- **Amazon (AMZN)** entered a 17-year contract with **Talen Energy Corp. (TLN)** to power its data centers with nuclear energy. It's also among a consortium investing \$500 million into SMR startup X-energy.

- **Meta Platforms (META)** inked a 20-year deal to buy power from **Constellation Energy Corp.'s (CEG)** Clinton nuclear plant in Illinois. The deal ensures the Illinois plant will stay operational past its previously scheduled 2027 closing.

A few years ago, I wrote we'd soon see nuclear-powered AI factories. It sounded a little kooky at the time. Now, it's happening!

We're entering the second atomic age. So, how do we profit from it?

We make money the *Disruption Investor* way.

Uranium producer Cameco has been on a tear since we first wrote about it. But Cameco isn't a great business and never will be. It sells uranium, a commodity. That means its fate is tied to the price of uranium, which it doesn't control.

In the past decade, uranium prices have traded as low as \$18 and as high as \$100. That's like running a lemonade stand where the price of lemons changes by 80% every few years. No matter how well you run the business, your margins are at the mercy of the market.

The way we think about commodity producers is simple: They're trades, not compounding machines. There's a time to own them, when the cycle turns in your favor. But we don't confuse a cyclical upswing with durable quality.

At *Disruption Investor*, we want to own companies that benefit from long-term, structural tailwinds. Businesses with pricing power and the ability to grow earnings regardless of where the commodity winds blow.

Cameco will rise and fall with uranium prices. But we want to own the picks and shovels. That's where the real compounding lives.

Let me hand it over to Chris, who's found what might be...

The best business in nuclear you've never heard of

What if I (Chris Wood) told you:

- More than 95% of all the world's nuclear power plants rely on critical technology provided by one company most folks have never heard of...
- These plants—and other nuclear installations like fuel mining and fabrication facilities—need this company's products and services from the time their construction begins throughout their decades of operation and eventual decommissioning and dismantling...
- And this company's tech is just as important for the advanced nuclear reactors being developed today—like small modular reactors (SMRs) and microreactors—as it is for conventional nuclear facilities?

It's true.

In fact, this company I'm referring to is one of the key linchpins holding the entire nuclear power industry together. It's arguably the best-positioned company in the space to profit from the nuclear renaissance.

Here's the story:

As Stephen talked about earlier, nuclear power is one of the safest sources of energy today—on par with solar and wind.

But the only reason it's safe is because of an ecosystem of experts and technologies that make it so.

Nuclear power plants are built with multiple layers of safety to contain radiation—like the thick concrete and steel containment building around the reactor as well as a series of emergency backup systems. But they still need a lot of ongoing support from the safety side of things.

For example:

- Companies like FoxGuard Solutions and Dragos protect plants from cyber-attacks with cybersecurity systems specifically designed for nuclear facilities.
- Companies like **Honeywell International (HON)** and **Johnson Controls International PLC (JCI)** provide fire protection and security systems.
- Companies like Jensen Hughes and **Tetra Tech (TTEK)** provide ongoing safety consulting and risk assessment.
- Companies like Platom and SECTOR provide specialized safety and regulatory compliance services.

The most critical part of this nuclear safety support sphere is radiation detection, measurement, and monitoring. Without it, there is no nuclear power. It's just as important as the fuel that powers the reactors. Because the ionizing radiation emitted by that fuel (and the material used to make it) is potentially so dangerous.

And when it comes to radiation detection, measurement, and monitoring, the most critical company is **Mirion Technologies (MIR)**.

Mirion provides the “eyes, ears, and seatbelts” that let any nuclear power project—whether it's a massive conventional plant from the 1970s or tomorrow's advanced small modular reactors (SMRs) and microreactors—see what the reactor is doing, keep people and the environment safe, and prove to regulators that everything is under control.

While CEO Thomas Logan officially founded Mirion in 2005 through the merger of three component businesses, its predecessor companies have been providing radiation safety products and services for more than 60 years.

Mirion is a global leader in 17 of the 19 major product categories it serves

For example, Mirion is a top provider of **reactor instrumentation & control technology**. It makes the neutron flux detectors and other sensors that measure conditions both inside and outside the reactor core and monitor power levels in real-time, critical safety functions for all reactors. These products are installed during plant construction and replaced or upgraded regularly throughout its life.

Mirion also supplies the hardened electrical assemblies that carry signals from these sensors through a reactor's containment structure—a vital component of any nuclear plant's control system.

In a recent high-profile project, the Bill Gates founded SMR startup TerraPower selected Mirion to design and fabricate a custom Nuclear Instrumentation System. The system—which highlights Mirion's ability to engineer state-of-the-art instrumentation for cutting-edge nuclear reactors—will continuously monitor the reactor's conditions and protect the core from getting too excited and overheating.

Mirion is also a leader in fixed and portable **radiation monitoring systems (RMS)** that continuously track radiation levels throughout a plant to protect workers and the public. These systems—typically installed during plant construction and replaced or upgraded during its life—work like smoke alarms for radiation. They constantly “sniff” the air, water, steam, and pipes and sound an alarm if radioactivity increases.

TerraPower also recently chose Mirion to provide an RMS for its Sodium reactor demonstration to track plant radiation and support safe operations.

This sort of monitoring is required by regulators throughout the world, so many of Mirion's solutions are mandatory for plants to comply with strict safety standards.

Protecting plant personnel from radiation is another area where Mirion is deeply involved. The company offers **dosimetry devices and services** that track how much radiation exposure individual workers receive to help ensure their safety.

Through its Imaging & Sensing Technology group, Mirion is also a global leader in **radiation-tolerant cameras and remote inspection systems**. These include Go-Pro-like cameras that can operate under extreme radiation and high temperatures, or underwater, to inspect reactor cores and other components during operation or maintenance without exposing personnel to radiation.

It's important to note that Mirion's technology extends beyond plant operations to **nuclear fuel handling, waste management, and decommissioning**.

For example, the company is providing the Burn Up Measurement System (BUMS) for X-energy's advanced SMRs. The system uses special detectors to ensure the reactor's innovative fuel is utilized to the fullest extent possible before disposal. By maximizing fuel burn-up and monitoring isotopic content, Mirion's system helps minimize waste and optimize fuel use.

The company also offers control systems and monitoring instruments needed for fuel storage, transport, and disposal.

Even when nuclear plants shut down, Mirion's solutions are in high demand. Decommissioning work requires extensive radiation measurement and safety controls—from mapping radioactive contamination in plant structures, to monitoring worker doses as they dismantle equipment, to long-term environmental surveillance at the site. Mirion's instruments and dosimetry badges are standard tools for the contractors performing all this work.

In other words, whether nuclear power plants are just starting up, operating in full swing, temporarily offline for refueling, maintenance & repair, or shutting down for good, Mirion finds business opportunities throughout their lifecycles.

This “cradle-to-grave” involvement in nuclear projects, which can have 100-year lifecycles, is a key aspect of Mirion's value proposition.

According to industry analysts, Mirion effectively occupies a “unique category-of-one” by being the only pure-play radiation safety provider offering such a comprehensive platform of solutions.

As I alluded to earlier, over 95% of the world's 450 or so existing commercial nuclear power plants already rely on Mirion's tech.

And Mirion is actively partnering with the leading SMR and microreactor developers to provide the critical radiation detection, measurement, and monitoring solutions these advanced reactors require.

In addition to the Nuclear Instrumentation System and the RMS Mirion is providing to TerraPower and the BUMS it's providing to X-energy, the company is working with a number of other SMR and microreactor developers. This widespread engagement indicates Mirion will likely be a ubiquitous behind-the-scenes partner for the advanced reactors in development as they progress from design to construction.

Whether it's a domestic SMR (or microreactor) rollout or an international project, Mirion's decades of know-how make it the go-to supplier for ensuring these new reactors operate smoothly and safely.

Meanwhile, not only do SMRs promise an increase in the number of reactors to help fuel Mirion's growth, they also tend to demand more instrumentation and safety equipment on a per-megawatt basis. Mirion's management has noted that the revenue opportunity per MW from SMRs is about 60% higher than it is for traditional large reactors.

In other words, the industry shift toward SMRs and microreactors plays directly to Mirion's strengths.

It's worth noting that Mirion's business is already seeing tangible growth from the trend. In the most recent quarter (Q1 2025) the company's nuclear power segment revenue grew 17.6% year-over-year. The increase was driven by both "the current installed base" and "new builds."

Because Mirion offers the "mission critical, compulsory solutions" needed for nuclear facilities to operate safely in highly regulated environments, the company has a resilient moat-like position as the nuclear renaissance takes hold. Once a customer installs Mirion's products, ongoing service, calibration, replacement, and eventual upgrades provide the company with substantial recurring revenue streams.

Mirion's reach extends beyond commercial nuclear power

The company also provides radiation safety solutions to 23 out of 31 NATO militaries as well as research labs.

Everything we've talked about so far is part of Mirion's "Nuclear & Safety" business. This segment accounted for 66% of total revenue in the first quarter of 2025.

The remaining 34% of revenue came from the company's "Medical" business. This segment is focused on improving cancer care by leveraging Mirion's expertise in ionizing radiation. The company provides hardware, software, and services for diagnostic imaging, radiotherapy, and nuclear medicine to hospitals, over 80% of cancer centers worldwide, and smaller healthcare providers.

Radiotherapy is the traditional type of radiation treatment you're familiar with. It's like shining a precisely-aimed spotlight from outside of the patient to kill cancer cells. Nuclear medicine is more like swallowing or injecting a tiny "radioactive smart pill" that travels through the bloodstream, homes in on cancer cells, and emits radiation from inside the patient to kill cancer cells.

Mirion's medical solutions—like radiation detection equipment, dosimetry services, and radiation therapy quality assurance—essentially help ensure these treatments are as safe for patients and providers as possible.

Put it all together and there's more than a lot to like about Mirion.

The company's Nuclear & Safety business competes with divisions of big companies like Thermo Fisher Scientific (TMO) and Baker Hughes (BKR) as well as a few focused specialists like Ludlum Measurements but none match Mirion's combined scope and focus in radiation safety.

In 2024, Mirion's total revenue of \$860.8 million reflected annual growth of 7.5%. I expect about the same rate this year, with an acceleration in growth beginning in 2026 as the nuclear renaissance starts to take hold.

Meanwhile, Mirion generated Real Cash Flow of \$68.8 million in the past year, an increase of more than 90% from the year before. I expect this figure to grow substantially in the coming years.

With a market cap right around \$5 billion today, I wouldn't call Mirion cheap in traditional value-investing terms. But considering the acceleration in growth I expect over the next few years, I'd still say the stock is attractively priced today.

Action to take: Buy Mirion Technologies (MIR) at current market prices and set a 35% hard stop.

Several of our holdings will report earnings before the next *Disruption Investor* publishes on August 7. Here's a rundown on the upcoming earnings releases:

- **Dynatrace (DT)** on 8/7
- **Mirion Technologies (MIR)** on 8/1
- **Nextracker (NXT)** on 8/1
- **Rockwell Automation (ROK)** on 8/7
- **Taiwan Semiconductor (TSM)** on 7/18
- **Tesla (TSLA)** on 7/23

Please note that the dates listed above are subject to change.

For the most up-to-date guidance on our current positions, [click here](#) to view the portfolio online.



Chris Wood and Stephen McBride

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